CV Intern Task: Egg Segmentation and Tracking

Code: https://github.com/rutvik-chandra-m1/egg-segmentation-tracking
Tracking video: https://drive.google.com/file/d/1Ne4JJu_ycORmcKbMRhbO92BvLx1tToE5/view?usp=sharing

Summary

- **Task 1.** SAM (ViT-B) masks for all 110 images. SAM has no notion of an egg, so an egg-likeness filter picked masks automatically on 52 images, and I reviewed every result and re-prompted the other 58 by hand.
- **Task 2.** A 88/22 split by physical scene to limit leakage. 40 logged trials led to a classical algorithm that fits ellipses to edge arcs and refines them with GrabCut.
- **Task 3.** Eval mean IoU 0.873 (95% CI 0.76 to 0.96), median 0.984, against 0.804 on dev. Eval beating dev reflects an easier scene mix, not better generalisation (section 3.1).
- **Task 4.** Both eggs tracked through hand carries and occlusion with 2 IDs and 0 identity errors on 64 hand-labelled checks.

1. Segmentation with a pretrained model

The dataset has 110 photos, stretched to 640 by 640, showing one to three eggs. I ran SAM's automatic mask generator with ViT-B weights on CPU, using a 10 by 10 prompt grid. That spacing of 64 px is fine for eggs that are all over 100 px wide, and halves the runtime of a 16 by 16 grid.

SAM is class-agnostic, so masks had to be chosen. A filter scored each candidate on ellipse fit, area, brightness and saturation. Its egg masks were accurate, but it also accepted background objects such as lids and keyboard keys, and found nothing on six images. Since these masks become ground truth, I reviewed every overlay:

Source of final mask	Images
Automatic selection, accepted on review	52
Hand-placed point prompts	52
Hand-drawn box prompts, where a point selected the whole pot	6

Known limits of this ground truth. SAM's own masks are weak on the three eggs under boiling water (124, 135, 145), whose edges are obscured by bubbles, and image 145 keeps a straight cut along the box prompt. Image 117, an egg behind a rod, is also rough. Eggs under leaves (10 to 13) follow the leaves. Scores on these images partly measure SAM's errors.

2. Classical segmentation

2.1 Dev and eval split

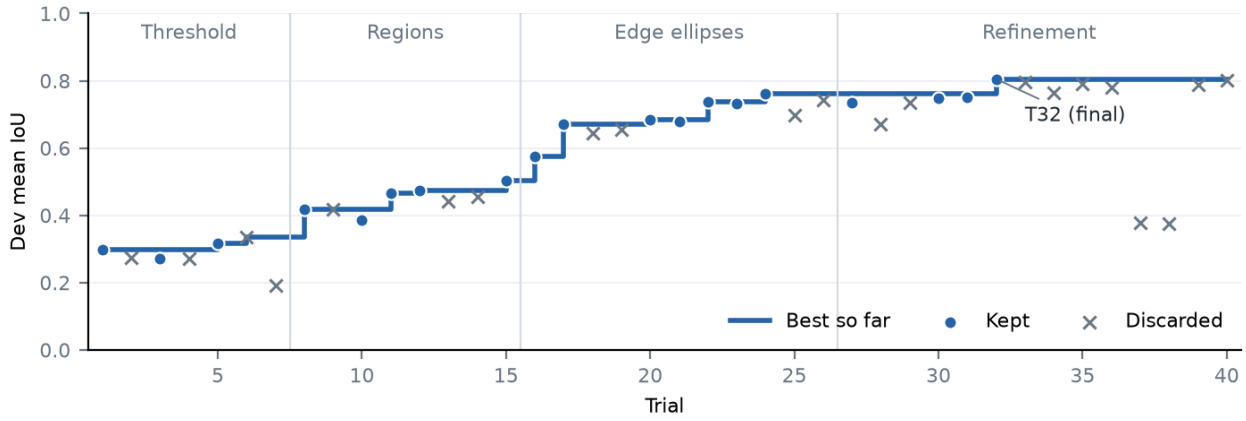
The photos are bursts of the same surface, eggs and light. Roboflow's own split and any per-image split would put near-copies on both sides, so I ignored it. I grouped images by eye into 20 physical scenes, such as the keyboard or the steel pot, and kept each scene whole. An exhaustive search chose the 22-image subset of scenes closest to the full set in multi-egg rate and egg size.

- **Result:** 88 dev images from 16 scenes, 22 eval images from 4 scenes (concrete, green carpet, keyboard, steel pot).
- **Check:** the closest dev and eval images are 18 apart in perceptual hash distance. Pairs within a scene go as low as 4, so no near-duplicates crossed.
- **Process:** every parameter was chosen on dev. Eval was run once, after the algorithm was frozen.
- **Limits:** the same physical eggs may appear in both splits. Eval has no 3-egg image, as nearly all of them are in the marble scene, which cannot be divided.

2.2 Final algorithm (T32)

- **Denoise and edges.** Median blur, then Canny with low thresholds (10, 30), so faint white-on-white outlines survive.
- **Arcs.** Split edge contours at sharp corners and keep the 80 longest smooth arcs.
- **Candidates.** Fit an ellipse to every arc and every pair of arcs, as an egg outline is often broken into pieces.
- **Score.** The fraction of the perimeter on a gradient normal to the ellipse, reduced for edges inside it and for colour inside it.
- **Select.** Take the best, then others scoring within 90% of it that do not overlap.
- **Refine.** GrabCut in a band around each ellipse, to follow the true outline and exclude fingers.

2.3 How the experiments led here



Thresholding (T01 to T07). Global and local thresholds on several channels. Best IoU 0.34. Brightness cannot separate white eggs from white surfaces.

Region proposals (T08 to T15). Threshold sweeps and Felzenszwalb regions, chosen by ellipse fit. Best IoU 0.50. T13 showed more proposals scored worse; visual review traced this to hole-filling and to eggs split along shading lines.

Edge-based ellipses (T16 to T26). Since outlines stay visible even where regions fail, ellipses are fitted to Canny arcs and ranked by perimeter gradient support, with interior penalties. Best IoU 0.76.

Refinement (T27 to T40). A stricter keep rule and GrabCut, which only helped once selection was reliable. T32 reached IoU 0.80. The last eight trials found nothing better, so tuning stopped.

Values are dev means against the SAM masks; each trial inherits from the one named in its parent column, and every trial can be rerun from `src/trials.json`. Repeat runs of T32 vary by about 0.001 because GrabCut is randomly initialised. Choosing the best of 40 trials makes dev scores slightly optimistic.

2.4 Experiment log

ID	Change and why	Parameters (from parent)	IoU	BF1	Count	PQ	Decision
T01	**Otsu on grayscale.** Baseline. Eggs are bright, so the simplest global threshold shows how far brightness alone goes.	channel=gray, proposals=otsu, selection=all	0.299	0.148	0.864	0.085	**Baseline.** Recall 0.87 at precision 0.32: bright pixels are mostly background.
T02	**Keep largest component.** T01 labels every bright pixel; most images hold one main object.	from T01: selection=largest	0.273	0.179	0.682	0.150	**Discarded.** IoU fell. The largest bright region is often the background, not the egg.
T03	**Median blur and open/close.** Speckle and texture fragment the mask and join it to the background.	from T02: denoise=median, open=7, close=7	0.272	0.185	0.693	0.149	**Kept.** Almost no IoU change, but median blur stayed useful. Morphology was dropped with region methods.
T04	**LAB lightness channel.** Perceptual lightness may separate eggs better than gray.	from T03: channel=l	0.271	0.179	0.648	0.148	**Discarded.** No gain over gray.
T05	**Whiteness channel (L minus chroma).** Hands, pink bowls and fabrics are bright but coloured; shells are not.	from T03: channel=whiteness	0.317	0.236	0.693	0.192	**Kept.** Best so far: colour rejects hands and fabric.
T06	**Adaptive threshold.** Uneven lighting defeats one global level; a local threshold might not.	from T05: proposals=adaptive	0.335	0.291	0.864	0.268	**Discarded.** Slightly higher IoU, but recall fell to 0.43.
T07	**Select by ellipse fit, not size.** The largest bright region is often the background; eggs are ellipses and backgrounds are not.	from T05: selection=egg	0.192	0.156	0.227	0.198	**Discarded.** Worst trial. One threshold rarely gives a clean ellipse. The idea was sound and came back in T08.
T08	**Threshold sweep proposals.** No single level suits every image, so try many and let the ellipse test choose.	from T07: proposals=sweep, sweep_levels=12	0.417	0.287	0.591	0.364	**Kept.** Largest gain so far: many levels give the ellipse test real choices.
T09	**Add dark-side sweep.** Against a white towel or bag the egg can be the darker region.	from T08: proposals=sweep, sweep_dark	0.418	0.287	0.614	0.364	**Discarded.** Unchanged IoU at 1.6 times the runtime.
T10	**Relax ellipse test to 0.75.** Fingers and shadows clip the outline, so 0.85 may reject true eggs. Many images currently return nothing.	from T08: min_ellipse_iou=0.75	0.386	0.277	0.773	0.308	**Kept.** IoU fell, but recall and HD95 improved greatly, as fewer images returned nothing.
T11	**Smoothness channel.** Worst images are white eggs on white textured cloth, which brightness can't separate but texture can.	from T10: channel=smooth	0.467	0.018	0.818	0.421	**Kept.** Solves white-on-white regions. Boundary F1 of 0.018 shows the filter window displaces the edge.
T12	**Whiteness + smoothness proposals.** The channels fail on different images, so pool their proposals and let the selector pick.	from T10: channel=whiteness, smooth	0.473	0.184	0.807	0.402	**Kept.** Best region result: channels fail on different images.
T13	**Add pale and gray channels.** More proposal sources for eggs that are neither bright nor on texture.	from T12: channel=whiteness, smooth, pale, gray	0.441	0.225	0.795	0.360	**Discarded.** More proposals scored worse, showing the selector, not proposals, was now the limit.
T14	**Felzenszwalb region proposals.** Graph-based regions give closed object shapes where no threshold isolates the egg.	from T13: proposals=sweep, felz	0.456	0.273	0.830	0.375	**Discarded.** Below T12 at about 7 times its runtime.
T15	**Stop filling holes, cap area at 0.45.** Visual review of T13: background sheets with an egg-shaped hole, once filled, pass the ellipse test.	from T12: fill_holes=False, max_area=0.45	0.504	0.190	0.727	0.462	**Superseded.** Fixed the hole-filling flaw found by visual review. Best region method, replaced by T16.
T16	**Edge-support selection on arcs.** Regions split eggs along shading lines, yet the outline is visible nearly everywhere. Fit ellipses to Canny arcs, rank by how much perimeter sits on a normal gradient.	selection=edge, denoise=median, proposals=, max_area=0.45, min_area=0.01	0.575	0.457	0.523	0.525	**Kept.** Change of approach. Boundary F1 more than doubled against T15.
T17	**Penalise edges inside the ellipse.** Ellipses fitted across textured background have support but a busy interior; a shell has a clean one.	from T16: interior_weight=4.0	0.672	0.514	0.614	0.609	**Kept.** Large gain: a clean interior separates shells from textured backgrounds.
T18	**Prefer larger ellipses.** Small fits on a shading line can out-score the whole egg.	from T17: size_weight=1.0	0.645	0.490	0.784	0.600	**Discarded.** Better counts, but lower IoU: review showed it choosing tubs and bowls.
T19	**GrabCut boundary refinement.** Eggs are not perfect ellipses and fingers occlude them; snap to real edges.	from T18: refine=grabcut	0.656	0.570	0.784	0.617	**Discarded.** Better boundaries, but it amplified wrong selections. Revisited in T30.
T20	**Penalise colour inside the ellipse.** Review of T19: tubs and bowls are ellipses too, and T18's size bonus chose them. Shells are nearly colourless. Branches from T17, dropping the size bonus.	from T17: saturation_weight=1.5	0.683	0.518	0.716	0.633	**Kept.** Small IoU gain, larger count gain.

ID	Change and why	Parameters (from parent)	IoU	BF1	Count	PQ	Decision
T21	**Lower support threshold to 1.5.** White eggs on white surfaces have faint outlines that fall below 2.5 levels per pixel.	from T20: edge_thresh=1.5	0.680	0.513	0.705	0.643	**Inconclusive.** No gain alone. T26 later showed it does help with more arcs.
T22	**Lower Canny to (10, 30).** Faint outlines never become arcs, so no ellipse is even proposed.	from T20: canny=10, 30	0.737	0.550	0.795	0.692	**Kept.** Large gain: faint outlines now become arcs at all.
T23	**Both lower thresholds.** Combine T21 and T22.	from T21: canny=10, 30	0.731	0.551	0.773	0.699	**Kept.** Slightly below T22 alone, kept to test with more arcs.
T24	**80 arcs instead of 40.** Textured images produce many long arcs that crowd out the egg's.	from T23: max_arcs=80	0.762	0.567	0.705	0.737	**Kept.** Best IoU so far, though count accuracy fell.
T25	**Region proposals as extra ellipse seeds.** T12's sweep found eggs on some images where arcs fail, so offer those as candidates too.	from T23: region_seeds=True, proposals=sweep, channel=whiteness, smooth, fill_holes=False	0.697	0.440	0.705	0.661	**Discarded.** Worse: poor seeds out-scored good arcs.
T26	**Back to threshold 2.5 with 80 arcs.** T21 and T23 showed the 1.5 threshold does not help once Canny is lower, so drop it.	from T22: max_arcs=80	0.742	0.552	0.750	0.703	**Discarded.** Below T24, which disproved my T21 reading: the low threshold is worth keeping.
T27	**Keep ratio 0.75 to 0.9.** Count accuracy is the weakest metric: extra ellipses score within 75% of the true egg.	from T26: keep_ratio=0.9	0.735	0.545	0.818	0.705	**Kept.** Count accuracy up with little IoU cost.
T28	**Raise minimum score to 0.5.** Absolute floor on weak single detections.	from T27: min_score=0.5	0.672	0.506	0.795	0.654	**Discarded.** Lost real eggs: IoU and recall both fell.
T29	**Polish ellipses onto edges.** Boundary F1 is 0.57 with correct selections: arc fits are rough.	from T27: polish=True	0.736	0.548	0.818	0.706	**Discarded.** No gain for 47% more runtime.
T30	**GrabCut in a narrow band.** T19 GrabCut raised boundary F1, and selection is now right often enough to refine. Narrow band limits leaks.	from T29: refine=grabcut, grabcut_band=0.04	0.747	0.630	0.818	0.717	**Superseded.** GrabCut now helps, as selection is right more often. Beaten by T31.
T31	**GrabCut in a wide band.** Compare with T30: does a wider band recover occlusions or just leak?	from T29: refine=grabcut, grabcut_band=0.1	0.752	0.662	0.818	0.721	**Kept.** Best boundary F1 to this point: the wider band beat T30.
T32	**T24 thresholds, keep 0.9, wide GrabCut.** T24 had the best IoU and T31 the best count and boundary. Combine them, without polish, which added cost for no gain.	from T24: keep_ratio=0.9, refine=grabcut, grabcut_band=0.1	0.805	0.703	0.875	0.780	**FINAL.** Best IoU, boundary F1 and count of all trials. PQ tied with T40.
T33	**Keep ratio 0.8.** 0.9 costs recall on multi-egg images. Look for a middle ground.	from T32: keep_ratio=0.8	0.796	0.694	0.795	0.765	**Discarded.** Lower on every headline metric.
T34	**Stronger interior penalty (4 to 8).** The interior penalty gave the largest single gain in T17, and foil and tubs still win sometimes.	from T32: interior_weight=8.0	0.764	0.682	0.841	0.768	**Discarded.** Over-penalises eggs with ash or speckle.
T35	**Stronger colour penalty (1.5 to 2.5).** Same reasoning for colourful containers.	from T32: saturation_weight=2.5	0.792	0.695	0.864	0.774	**Discarded.** Worse overall, and backlit eggs 20 and 24 join the worst images.
T36	**Lower bar for same-size eggs.** Review of T32: 140 and 144 find 1 of 3 eggs. Eggs in a photo share a size and junk does not.	from T32: same_size_ratio=0.7	0.780	0.688	0.795	0.766	**Discarded.** Best recall (0.90), but count accuracy fell to 0.80.
T37	**Penalise uneven brightness inside.** Review of T32: bag-flap ellipses span a dark slit, an egg interior is even.	from T36: variance_weight=1.0	0.378	0.343	0.455	0.376	**Discarded.** Flawed test: penalty pushed real eggs under the fixed floor. Retested in T39.
T38	**Same-size ratio 0.6.** Check the sensitivity of the T36 value.	from T37: same_size_ratio=0.6	0.376	0.341	0.432	0.378	**Discarded.** Same flaw as T37, so uninformative.
T39	**Gentler brightness penalty, lower floor.** T37 failed because the penalty pushed true eggs below the fixed 0.35 floor, a flaw in the test. Retest fairly.	from T32: variance_weight=0.4, min_score=0.25	0.788	0.685	0.784	0.763	**Discarded.** Fair retest of T37: still below T32.
T40	**Same-size ratio 0.8.** T36 at 0.7 traded too much precision. Try a milder bar.	from T32: same_size_ratio=0.8	0.801	0.700	0.864	0.780	**Discarded.** No gain over T32.

IoU: mask overlap. BF1: boundary F1 at 3 px. Count: share of images with the correct number of eggs. PQ: panoptic quality, which also penalises merged or split eggs.

3. Evaluation and analysis

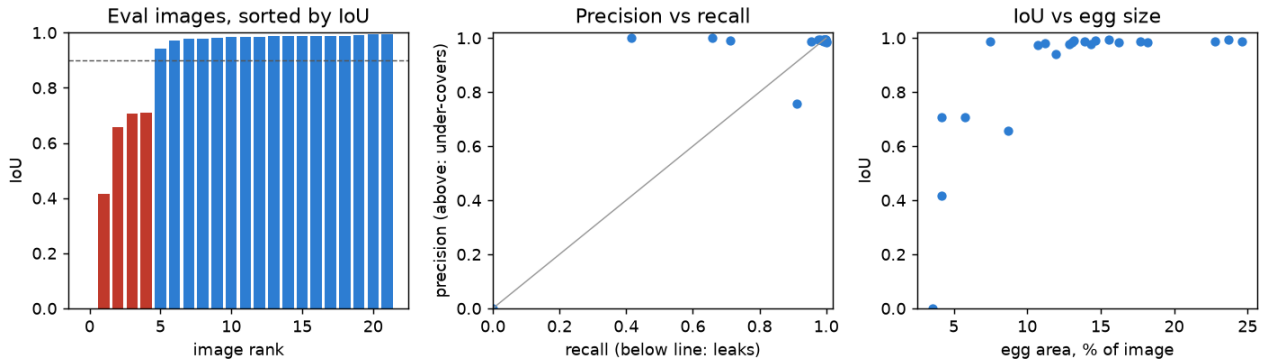
3.1 Quantitative results

Metric	Dev mean	Eval mean	Eval 95% CI
IoU	0.804	0.873	0.759 to 0.959
Dice	0.841	0.904	0.797 to 0.977
Precision	0.840	0.936	0.835 to 0.992
Recall	0.867	0.889	0.774 to 0.974
Boundary F1 (3 px)	0.702	0.790	0.651 to 0.910
HD95 (px, lower is better)	63.975	26.730	6.327 to 59.827
Panoptic quality	0.780	0.853	0.719 to 0.958
Correct egg count	0.875	1.000	1.000 to 1.000

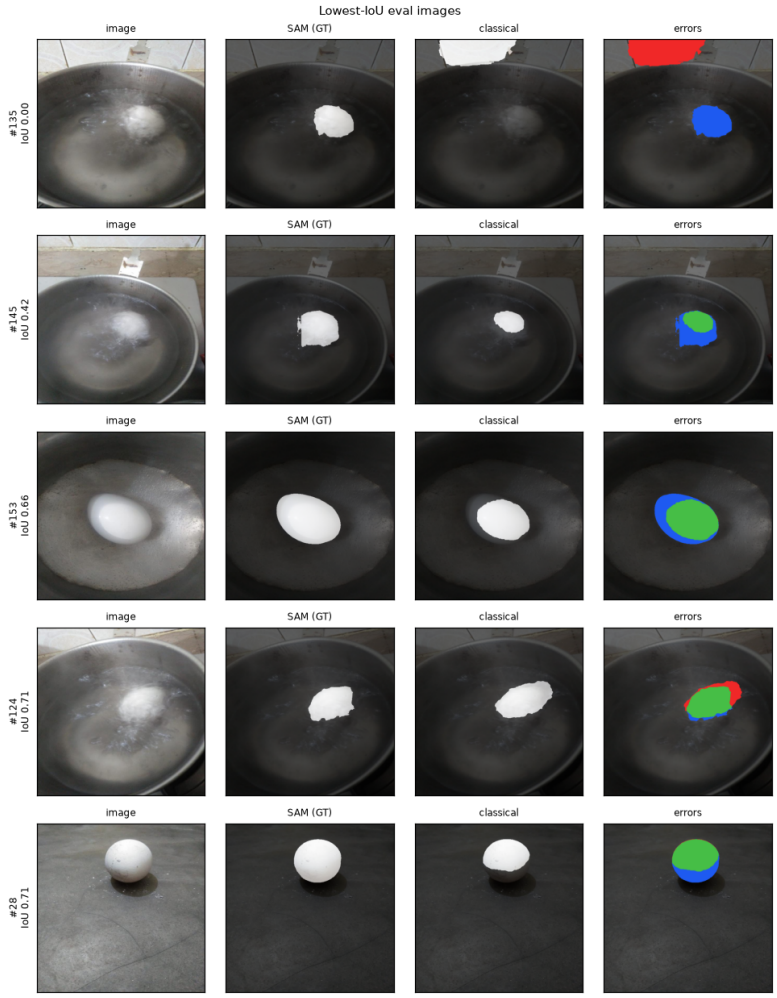
At instance level, with a match needing IoU above 0.5, eval has 25 true positives, 2 false positives and 2 false negatives (precision and recall 0.93). 17 of 22 eval images reach IoU 0.9 or more, and the median of 0.984 shows a typical image is near SAM's quality.

Read the eval mean with care. Eval beats dev (0.873 against 0.804), but this should not be read as generalising better. With 22 images from 4 scenes the interval is wide, and those scenes contain none of the white-on-white photos (bag, tray, towel) that cause most dev failures. The dev score is the safer guide to unseen scenes. Likewise, the 100% count accuracy on eval is limited by having no 3-egg images.

Scene	Images	IoU	Boundary F1	Precision	Recall
steel pot	10	0.762	0.620	0.872	0.787
concrete	5	0.929	0.874	0.988	0.940
green carpet	3	0.987	0.952	0.987	1.000
keyboard	4	0.991	0.987	0.993	0.998



3.2 What went wrong



Green: agreement. Red: false positive. Blue: missed.

- **Submerged eggs (135, 145, 124).** Bubbles destroy the outline the method depends on. In 135 (IoU 0.00) the best ellipse is a wall tile, the only wrong object on eval. These three images also have the weakest ground truth.
- **Shading splits (153, 28).** IoU 0.66 and 0.71. Precision is near 1 but recall low: the ellipse stops at the terminator between light and shadow. GrabCut then likely treats the darker part as background, so refinement does not recover it.
- **Small eggs.** Every eval image with IoU below 0.9 has an egg under about 9% of the image, while all larger eggs score above 0.9. Small outlines give fewer and shorter arcs. This overlaps with the causes above, as the pot images have small eggs.
- **On dev: white on white.** Many of the worst dev images (3, 5, 109, 123, 125) are white eggs on white cloth or bags, where the chosen ellipse can be a bag flap or nothing. Eval has no such scene, so this failure is hidden from its score.
- **On dev: multi-egg images.** The 0.9 keep ratio that removes junk also drops real eggs, as in 140 and 144, where one of three eggs was found.

3.3 Where to improve

- **Combine edge evidence with colour models.** A shadowed lower half fails GrabCut because it is darker. Normalising illumination first, for example dividing by a heavily blurred image, should reduce terminator cuts.
- **Use a shape prior during refinement.** Constraining GrabCut to stay near an ellipse would keep the whole egg in shading cases.

- **Replace the keep ratio with a learned threshold or a per-image model.** The same-size rule in T36 raised recall to 0.90, so a stricter version, such as also requiring a clean interior, may recover eggs without adding junk.
- **Evaluate on more scenes.** A larger eval set, or cross-validation by scene, would narrow the interval and cover white-on-white surfaces.

4. Video tracking

In the video, a hand carries each egg across the frame in turn, hiding it. Egg 1 goes right and comes back, then egg 2 goes right, so the eggs swap left-to-right order for a while and pass under the same hand. They look identical, so identity rests on motion and on reasoning about which egg is missing. I ran the final algorithm on each frame with three changes: a size range for this video, a lower keep ratio since two eggs are present, and no GrabCut, which is too slow per frame and unnecessary for tracking.

- **False detections.** The first run had 6 IDs for 2 eggs, from table reflections and the title cards. Both are well-shaped but not white, so strong detections must now pass an absolute whiteness test.
- **Motion.** A Kalman filter with constant velocity predicts each egg. Once an egg is hidden for over 3 frames its prediction stops at the last seen position.
- **Association in four stages.** Tracks match to strong detections, then unmatched tracks match to nearby white slivers, such as an egg showing between fingers. Remaining strong detections go to lost tracks, nearest first, and finally lost tracks can claim large white blobs.
- **New IDs.** Only made when no lost track can claim a detection that persists for 8 frames, because eggs do not appear from nowhere.

Result

Measure	Value
Unique egg IDs in the video	2
Keyframes checked, by egg	64
Correct identity	60
Wrong identity	0
No track, with the egg hidden in the hand	4

I labelled positions by hand on 32 keyframes from frames 470 to 995, covering all of the carries, and viewed frames 726 to 765 at three-frame steps, where both eggs are under the hand, to confirm the identities. Labels come from thumbnails, so position agreement uses a tolerance of 50 px.

Limits. This is a two-egg clip, and the reassignment rule relies on it: with several eggs hidden at once, identity would be a guess based on last position. The masks are ellipses on this video and may run slightly off while an egg is spinning. The whiteness threshold is tuned to this video and would need adjusting for others.



Every 15th frame from 470 to 995. Egg 1 goes right and returns by frame 695, then egg 2 goes right from frame 755.

5. Reproducing this work

Setup and run instructions are in the repository's README. Results here come from `outputs/`, regenerated by `src/build_report.py`.